

Package for Eco-friendly Pest Management of Tomato



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Foreword

Concern about environmental safety and sustainability of land productivity is increasing among the scientists, administrators and environmentalists. The strategy adopted during the green revolution era can not be valid any more under the prevailing conditions. A new strategy of living with the nature and nurturing it for sustainable high productivity should be evolved. Eco-friendly pest management shows us the way to effectively use the available natural resources for the benefit of the mankind. Due to the significant increase in the human population, and the consequent increase in the amounts of food and grains produced, many small-scale farmers adopted the use of pesticides as a means of pest control. Chemical control forms the prime and foremost method for the management of insect pests of agricultural and horticultural crops. Prolific use of chemical insecticides significantly curtailed the insect pests in the past but in due course it resulted in the development of resistance to insecticides in insects, environmental degradation and increase in the cost of cultivation. The use of pesticides is being done indiscriminately for production of different crops, fruits and vegetables, which contaminate them to a great extent, especially in vicinity of big cities. Almost five decades of pesticides use have left us at a tragic legacy; severe contamination of our soils and water system, increased cancers, birth defects and other ailments in humans and the emergence of powerful pests which are resistant to chemical pesticides. About 508 insect species have developed resistance against different chemicals. According to the World Health Organization (WHO) approximately one to two million persons are affected every year because of pesticides. Considering such ill effects of pesticides, both the scientists and planners realized the need to reorient the methodology and efforts to overcome these significant yield limiting factors with more eco-friendly manner. This Technical Bulletin “***Eco-friendly pest management practices for Tomato*** ” can provide *food security* by preventing and reducing crop losses, promote *self-reliance* by women farmer participatory approach, contribute to *poverty alleviation* by focusing on women farmers and *protect environment* and *health* by reducing chemical inputs on our unique planet earth.

Authors

India is the world's second largest producer of vegetables next to China. Orissa is the third leading state of India in the production of vegetables followed by West Bengal and Uttar Pradesh. According to the recommendations given by Indian Council of Medical Research (ICMR) an average man with vegetarian or non-vegetarian food habit should consume 125 gram of leafy vegetables, 100 gram of roots and tubers and 75 grams other vegetables. The recommendations for an average woman are more or less same with exception in roots and tubers which should be consumed @ 75 gram per day. Vegetables are the only natural sources of protective food, supplying all the nutrients specially minerals, vitamins and crude fibre. Tomatoes are used directly as raw vegetables in sandwiches, salad etc. Several processed items like paste, puree, syrup, juice, ketchup, drinks, whole peeled tomato etc. are prepared on a large scale. Tomato is a very good appetizer and its soup is said to be a good remedy for patients suffering from constipation.

Botanically tomato is *Lycopersicon esculentum* Mill., belongs to family *Solanaceae*. It is a herbaceous, annual to perennial, prostrate and sexually (occasionally asexually) propagated plant. Tomato is grown in almost all the states in India. Bihar, Karnataka, Orissa, Maharashtra, Andhra Pradesh, Madhya Pradesh, Assam, Tamilnadu, Gujarat, West Bengal, Haryana and Punjab are the largest producer of tomato in the country. The production of tomatoes in India got a boost when Pepsi promoted its cultivation in Punjab with its high yielding seeds.

Now F_1 hybrids of tomato are being widely used in India in all the important tomato growing states because of their increased availability. Maharashtra Hybrid Seeds, Indo-American Hybrid Seeds and Proagro are major producers of these seeds in India. At present, Karnataka,

Maharashtra, Punjab, Haryana, West Bengal, Orissa and Uttar Pradesh account for maximum area under hybrid tomatoes.

The Scientists of Indian Institute of Horticultural Research, Hessarghatta, have developed two new varieties of hybrid tomato, 'Arka Shresta' and 'Arka Abhijit, which are resistant to bacterial wilt. The average yield of these varieties is about 45 tonnes and 55 tonnes per hectare respectively.

In tomato processing, tomato seed is obtained as a byproduct. It constitutes about 0.5 per cent by weight of tomato. The seed contains edible oil, which can be extracted. The potential value of this oil in the country is estimated at Rs. 50 crores annually although; tapping it to this extent is not possible. The Oil Technological Research Institute Anantpur has developed technology for the extraction of this oil.

NUTRITIONAL INFORMATION

COMPONENT	PER 100 GRAM EDIBLE PORTION
Food energy	22.0 kilo calories
Moisture	93.5 percent
Protein	1.1 gm
Fat	0.2 gm
Carbohydrates	4.7 gm
Calcium	13.0 mg
Phosphorous	27.0 mg
Iron	0.5 mg
Vitamin A	900 International Unit
Thiamin	0.06 mg
Riboflavin	0.04 mg
Niacin	0.7 mg

Ascorbic acid	23.0 mg
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From the dawn of the history of mankind, we know that our natural environment has been the source of our progress and development. Although, science and technology has progressed far, human beings are now subjected to live in a much polluted environment and unhealthy conditions. Unfortunately, our environment is deteriorating day by day, until one day; it will become a threat to human existence. Today, the main causes of human deaths are cancer and diseases of the heart and brain. Surveys show that 75% of cancer is caused by environmental pollution and various carcinogenic chemicals absorbed by humans. We are the ones who are continuously destroying our very own living environment. Man cannot live without even one of the essential elements of life-air, soil, water and sunlight. It should be realized that humans need to co-exist in harmony with natural environment in order to survive.

Several types of insects attack the plant and fruit, which can also be afflicted with diseases. Tomato fruit borer (*Helicoverpa armigera*) and serpentine leaf miner (*Liriomyza trifolii*) are some of the important pests of Tomato. For quality production of tomato serious attention must be paid to empower the farmwomen and farming society for Eco-friendly pest management, which is the most important concern of the whole world today.

Eco-friendly pest management package for Tomato

1. Collection and destruction of crop debris of previous harvested crop.

2. Deep ploughing in summer season helps in exposing resting stages of pests and soil borne nematodes to sunlight.
3. At the time of last ploughing incorporate Karanja (*Pongamia glabra*) oil cake @ 250 kg/ha in the soil. The pesticidal effect of Karanja controls 30-40% nematodes which causes wilt and also provides nutrient to the crop.
4. Collection and destruction of egg masses, larvae of fruit borer and tobacco cutworm etc.
5. Raised nursery beds at least 10 cm for good drainage thereby avoid damping off in Solanaceous nurseries by preventing soil borne fungi i.e. *Pythium*, *Phytophthora* and *Rhizoctonia* etc.
6. Seed treatment with *Trichoderma viridae* @ 4g or Carbendazim @ 2g /1000 g of seeds to prevent soil borne fungal infection.
7. The seeds of tomato are soaked in 3 day old butter milk for 6 hours before seeding. This results in 80% germination of the seeds.
8. Use of 100-mesh nylon net in nursery beds to avoid entry of whitefly *Bemisia tabaci* and transmission of leaf curl disease in tomato.
9. Soaking of seeds of tomato for 6 hours in 20 ml extract of Kohithapa (*Bougainvillea glabra*) with 1 liter of water is found effective to control damping off disease.
10. Drenching of hukka water, once in a week keeps various insect pests away from the nursery. Spraying of hukka water is also effective to control the pests on transplanted seedlings. Hukka water contains nicotine sulphate which acts as controlling substance for caterpillars and other pests.

11. Tender bamboo shoot is cut into small pieces and kept in a pot with water. After 2- 3 days, it is sprayed on the crop. After 15 days of transplanting it is applied at 15 days interval and about 100 liters is used/ ha. These 10 kg tender bamboo shoot are used for 100 liters of water for the control of different pests.
12. Root rot and damping off in tomato is controlled by applying *Cynadon* leaf extract. Fresh leaves of *Cynadon dactylon* are dried in shade. The dried sample is powdered and then mixed with water and placed in air tight mud pot and left undisturbed for 24 hours. This mixture is filtered twice, first using gunny cloth and then with a white piece of cloth. The clear mixture is ready for use. One liter of this mixture is added to one liter water and sprayed @ 10 liter / acre, when tomato seedlings are 15 days old .This mixture can be sprayed at frequent intervals until harvest.
13. Root knot nematodes cause” galls” and reniform nematodes cause “dirty roots”. Their presence can be detected by “Trypan blue” stain which turns nematode eggs sac dark blue whereas, roots stay uncoloured. Soil solarization after irrigation with 60-100 gauge black polythene sheets in nursery beds for about 15-21 days, helps in killing weed seeds, nematodes, and resting stages of insects and diseases.
14. Field sanitation and removing alternate hosts helps the crop free from pests.
15. Application of Neem cake / Pongamia cake @ 250 kg/ha at the time of planting for the reduction of root knot

nematode, leaf miner and fruit borer egg laying, termite infection and spotted wilt disease. Repeat cake application at flowering to reduce incidence of fruit borer.

16. Destruction of affected plant / plant parts and light irrigation reduces late blight.
17. Installation of pheromone traps for Fruit borer and Tobacco caterpillar @ 12 traps /ha with replacement of lures once in 15 days. Trap should be installed in the field in such a way that the position of the lure is always 2 to 3 feet above the crop canopy. The trapped moth should be collected and killed daily. ETL for fruit borer is 2 moths /day /trap.
18. Conserve parasitoids like *Trichogrammatoidae armigera*, *Compoletis chloridae*, *Carcella illota*, *Eriborus argentiopilosus*, *Hyposoter didynotor* , *Palexorista laxa* , *Goniophthalmus halli* for *H.armigera*; *Telenomus remus*, *Chelonus heliopae* , *C. formosanus*, *C. blackburni* , *Eriborus sp.*, *Cotesia colemani*, *Cotesia marginiventris*, *Microplitis sp.*, *Charops obtuse*, *Euplectrus sp.*, *Peribae orbata* for *Spodoptera litura* ; *Pristomerus testaceous*, *Cremastus flavoorbitalis*, *Bracon greeni*, *Phanerotoma sp.*, *Campyloneura sp.* for *Lorbanalis*.
19. Avoid excess dose of Nitrogen.
20. Spray 4% NSKE (40 ml/ liter) (Flow chart for the preparation of NSKE is given in the last of the text) or neem seed powder (7%) at 15 and 25 days after planting and it should be repeated for 2 times at 10 days interval for the control of fruit borer and serpentine leaf miner, if required.

21. Interplanting with beans reduces the attack of serpentine leaf miner.
22. Roguing of affected plants particularly virus-infected plant for the control of Leaf curl disease. Yellow plastic mulch is found effective to reduce the incidence of leaf curl virus in tomato.
23. Release *Trichogramma priteosum* @ 50000/ha at pre flowering and to be continued for 6 times at an interval of 7 days.
24. Marigold raised on bunds of tomato crop in mid hills of Chennai lessened the insect pest and diseases in tomato crop. Intercropping of marigold has also been found effective for the control of wilt in tomato.
25. The seed extract of Kochilla (*Strychnos nux-vomica*) can control tomato fruit borer (*Helicoverpa armigera*) within 48 hours when tomato pieces treated over 60 minutes and fed to borers. Soil application of seed extract of Kochilla @ 1.5 g /plant twice at 20 days interval is efficient in controlling fruit borer of tomato.
26. Installation of Yellow pan /sticky traps 15 cm above the canopy against white fly and thrips etc. @ 10 trap/ha. Locally available empty tins can be painted yellow/ coated with grease / vaseline/castor oil on outer surface may also be used as yellow sticky trap.
27. TRAP CROP TECHNOLOGY: Tomato fruit borer is the major pest on developing fruits and is responsible for major yield loss in tomato. Intercropping a tall variety of marigold as trap crop in a row after every 16 rows of tomato is very effective.

Plant 45-day old marigold seedlings and 25-day old tomato seedlings to synchronize flowering in both the crops. Maximum egg laying is observed on marigold flowers and the movement of larvae from marigold to tomato is not significant. Eggs and larvae are removed from field along with the flowers. This trap cropping system also helps in reducing the root knot nematode infestation.

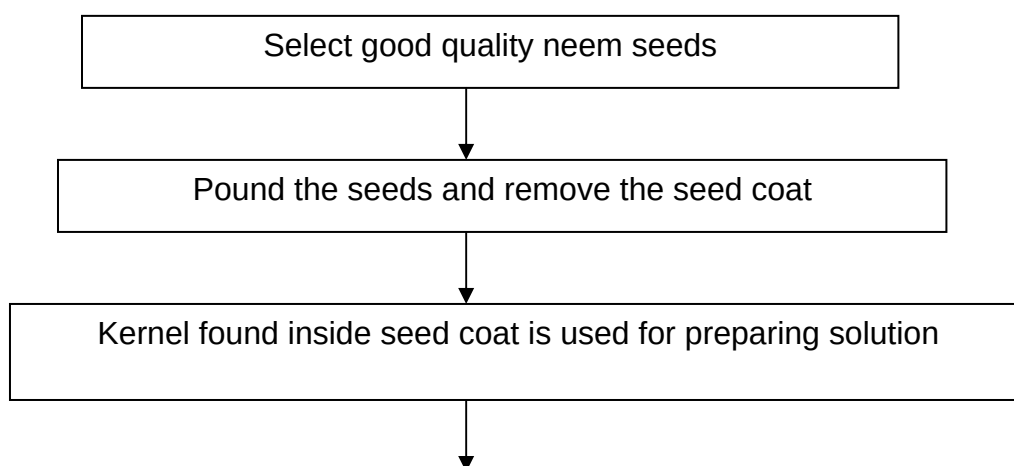
28. Install bird percher @ 50/ha to encourage birds to feed on pests.
29. Spray *Bacillus thuringiensis* var *kurstaki* @ 500g/ha against lepidopteron pests.
30. Spray NPV 250 LE four times in the evening at an interval of 4-7 days for a pure tomato crop. Flow chart for the preparation of NPV is given in the last of the text. If marigold is grown as a trap crop, spray it only twice at 28 and 35 days after planting.
31. Dig a pit of required size in barren field itself to mix cow dung, cow urine, chilli and garlic in sufficient water. These are kept in the pit for at least 10-15 days for proper decomposition. Then the decomposed mixture is sprayed in Tomato at 15 days interval.
32. To control wilt in tomato, a mixture is prepared by mixing 1 g hing (*asfoetida*) and 10 g turmeric powder in 10 litres water. The seedlings of tomato are kept in the solution for 30 minutes before transplanting. By using this method wilt can be controlled up to 83 %.
33. About 15-20 g turmeric powder is put into 1-1.5 liters of water to make a solution. The solution is sprayed over the affected

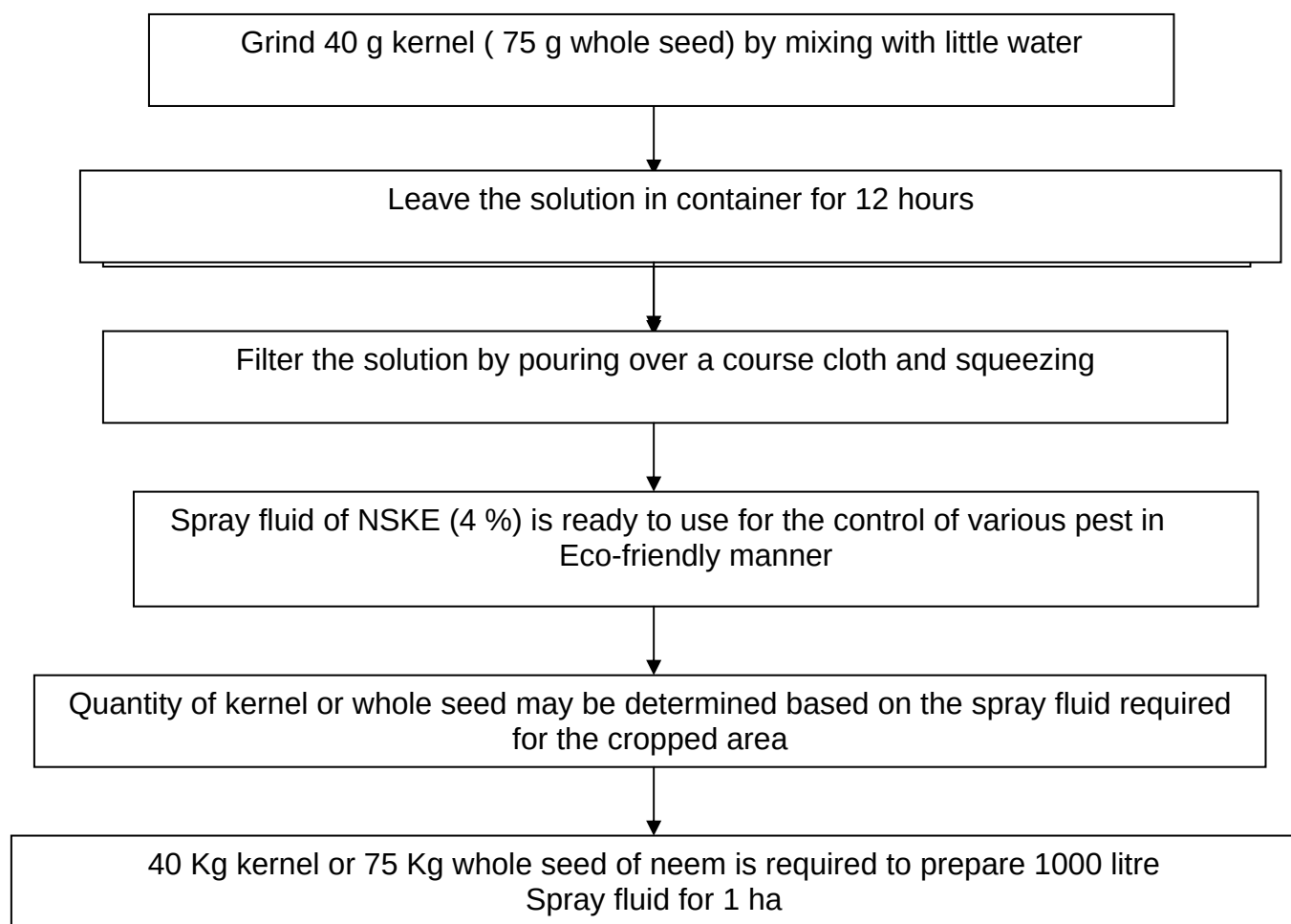
crop. The anti fungal property of turmeric kills the fungus causing wilt in tomato.

34. Tomato is prone to mosaic virus. In order to control the virus application of powdered form of neem cake/ mustard cake with water in root zone of the crop is effective. Spraying of hing solution is also found effective to control mosaic virus.
35. Application of tank silt @ 25 tonnes /ha not only supplies micronutrients that build resistance to pests but also save cost on plant protection.
36. It is a common practice to sprinkle wood ash in kitchen garden on vegetable crops, spread it around plants to ward off pests and to enhance nutrients status of the soil. Wood ash@ 20 kg/acre is dusted with hand or duster on seedlings in the nursery as well as on the young seedlings, which protect the plant from pest and disease attack. Wood ash is a source of phosphorus for plants and it also acts as a physical poison causing abrasion of epicuticular waxes and thus exposing pests through desiccation.
37. Custard apple (fruits) is soaked in water for 12 hours and this solution is mixed in 1 liter water. Spraying is done on affected plants for the control of different pests.
38. Tobacco leaves are soaked in water for 24 hours and boiled in 1 liter water. About 10-15 ml of this solution is mixed with 1 liter water and spraying is done on affected plants for the control of wilt.
39. Aphids and mites can be controlled by broadcasting ash of *ashan* wood.
40. Rice starch and animal urine are spread for the control of biting and chewing type insects.

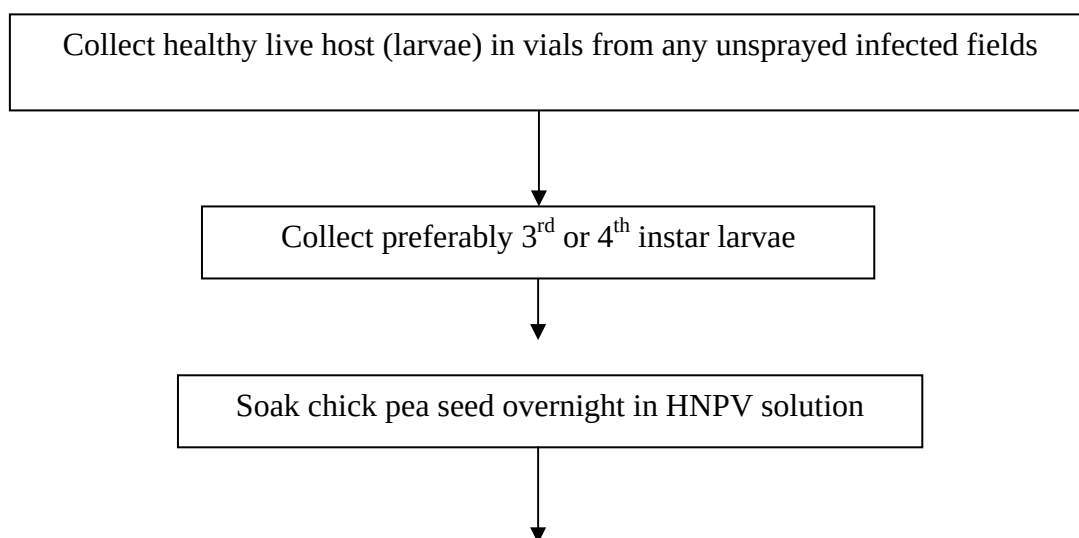
41. Regular destruction of damaged fruits at each harvest.
42. Use of coconut dust and wood ash in vegetable seeds, particularly in tomato helps to save seeds from pest / disease damage in storage.
43. For the safe storing of tomato seed from storage pests, seeds are dried under sunlight after peeling of tomato and mixed with rice husk and then the seeds are kept in earthen pots. The opening of pot is covered with cow dung. Dry seeds are kept above 2.5 -3 m from the *chulha* under moisture free condition.
44. Seeds of tomato may be stored in bamboo up to one year after wrapping in *shevni* leaves.
45. For storing the seed of tomato for one year, 2 kg tomato pulp is mixed with 2 kg fine soil. Small round shaped balls are made and hanged in the kitchen in the form of a garland with the help of a thin rope.

**FLOW CHART FOR THE PREPARATION OF 4% NEEM SEED
KERNEL EXTRACTS (NSKE) FOR ECO-FRIENDLY PEST
MANAGEMENT**





FLOW CHART FOR THE PREPARATION OF NPV



Feed to the larvae in vials @ 2-3 seeds/ larvae



Wait 4-5 days and allow the infected larvae to die naturally



Collect larvae as soon after death in a Separate container without causing any damage



Keep the container in refrigerator for need based further processing



Blend the dead larvae in suitable blender to crush the larvae tissue and release the NPV



Collect the material by washing with distilled water



Filter the material through muslin cloth to remove larvae body parts



Centrifuge the filtered solution for one minute at 500 RPM



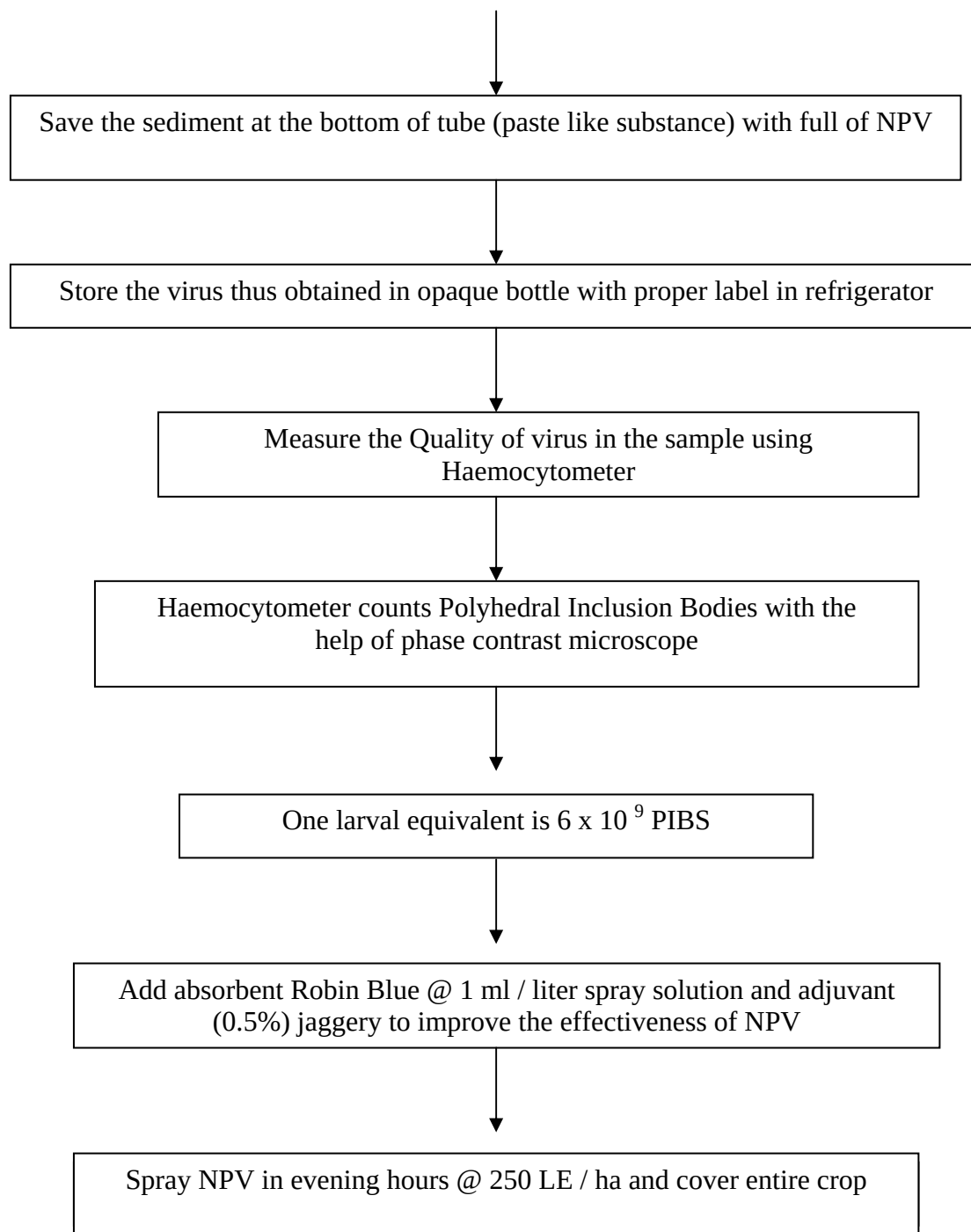
Discard pallet as it contains tissues and debris



Centrifuge supernatant solution for 15 minute at 3000 to 5000 RPM to separate virus from homogenous fluid



Poured off the liquid which is turbid and loose on the top of the tube



The National Research Centre for Women in Agriculture (NRCWA), established in 1996 under the aegis of Indian Council of Agricultural Research (ICAR) with its headquarters at Bhubaneswar, Orissa and sub-

centre at Bhopal, Madhya Pradesh, in India. The present mandate of the center is “To identify gender issues and test appropriateness of available farm-technologies/ programmes / policies with women perspective for promoting gender mainstreaming in research and extension for empowerment of farmwomen and capacity building of scientists, planners and policy makers to respond to the needs of the farm women.”

For any information please write to:

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